Lab Procedure
Encoders

Introduction

Ensure the following:

1. You have reviewed the **Application Guide – Encoders**
2. Make sure the **Mechatronic Sensors Trainer** is out of its case; it is powered using the provided USB cable connected to the PC.
    a. The LED next to the USB C port will stay white after the touch screen starts loading.
    b. The load screen with the Quanser Logo should disappear after a few seconds and you should see some evenly spaced crosses on a black background on the touch screen.

Exploring Decoding Logic

1. Launch Visual Studio Code or your preferred Python IDE. If using Visual Studio Code, click on **File** > **Open Folder**…, and browse to the directory containing the python files and lab procedure for the Encoders lab (…/sensors_trainer/1_fundamentals/encoder/hardware/python). Open this directory.
2. Open **encoder.py**. Make sure that on the initialization line `knobEncQuad` is set to 1 which means that the encoder will be working in X1 decoding:

    ```
    with SensorsTrainer(knobEncQuad=1) as sensors:
    ```

3. Turn the encoder knob to a position where the line is fully horizontal or vertical (could be pointing up/down/left/right) to facilitate recognizing when a full rotation happens.
4. Run **encoder.py** using the ▷ button or the terminal. This will open an **`Encoder Signals`** plot showing encoder pulses on the top plot and encoder counts on the bottom plot .
5. Look at the plot of encoder pulses. Turn the encoder one full turn clockwise, how many counts does the encoder read? Does turning clockwise make the number increase or decrease?
6. Bring the encoder counts back to 0. Repeat the last step but turning counterclockwise.
7. Very slowly turn the encoder knob while observing both scopes. When do the encoder pulses trigger a change in encoder counts? Consider the rising and/or falling edges in both the A and B signals.

8. Since the counts are proportional to the angle of the knob, how would you use the encoder counts (with the encoder set to X1 decoding) to figure out the angle of the encoder knob in degrees and radians? What is the minimum angle you can read (resolution)?

9. Click back on the terminal in Visual Studio Code and stop the script using **`Ctrl+C`** and clicking enter when prompted.

10. Find the section of the code with the comment (**`# Output encoder counts and angle every half second`**), replace the **`countsPerRev`** with the number of counts you get in one full turn of the encoder. Add a print so every half second you can see a printed output of the encoder counts, and the angle of the encoder knob in both radians and degrees.

11. Run **encoder.py** using the button or the terminal.

12. Turn the encoder in both directions and take a screenshot of your print outputs in the terminal.

13. Click back on the terminal in Visual Studio Code and stop the script using **`Ctrl+C`** and clicking enter when prompted.

14. In the initialization line **`with SensorsTrainer(knobEncQuad=1) as sensors:`** set **`knobEncQuad=2`**. This will set up the device to have 2X decoding.

15. Run steps 3 – 13 again but with X2 decoding.

16. In the initialization line **`with SensorsTrainer(knobEncQuad=1) as sensors:`** set **`knobEncQuad=4`**. This will set up the device to have X4 decoding. Note that **`knobEncQuad=4`** is the default value of that argument, so even when initializing without that argument (**`with SensorsTrainer() as sensors:`**), it will assume X4 decoding.

17. Run steps 3 – 13 again but with X4 decoding.

18. Open the Encoder's Datasheet located in the encoder folder (…/sensors_trainer/1_fundamentals/encoder). The encoder in the Mechatronics sensors board is the PEC12R-4225F-N0024. What is the resolution of the encoder based on the datasheet? Does the Quadrature Output Table match what you saw when rotating the encoder?
    Note that labels for encoder outputs (channels A and B) are assigned to match the standard across Quanser devices, which could lead to a mismatch with the datasheet. The standard in Quanser devices is B leads A for a CW direction.

19. Stop, save and close your program.

20. Power OFF the Mechatronic Sensors Trainer by disconnecting its USB C cable, disconnect any external sensors and pack them along the base and the USB cables in the provided case.